Week 8 Problem Set: The Derivative as a Function

Calculus Honors · Module 4: The Derivative · Lessons 4.03 and 4.04

Name: _____ Date: _____

> Show all of your work. A correct answer with no reasoning earns partial credit at most. Keep your writing neat and vertical, line up your equal signs, and label your final answers. Justify differentiability decisions with the one-sided difference quotients, not just a guess.

PART A · COMPUTING F'(X) FROM THE DEFINITION

1. Use the definition to find $f'(x)$ for $f(x) = x^2 + 5x$. Show the difference-quotient algebra in full, then evaluate $f'(0)$, $f'(-2)$, and $f'(3)$.

2. Use the definition to find $f'(x)$ for $f(x) = \sqrt{x+1}$ by rationalizing the numerator. State the domain on which your formula for $f'(x)$ is valid.

PART B · READING THE SIGN OF THE DERIVATIVE

3. The graph of g rises on $(-\infty, 1)$, has a horizontal tangent at $x = 1$, falls on $(1, 4)$, has a horizontal tangent at $x = 4$, and rises again on $(4, \infty)$. State the sign of $g'(x)$ (positive, negative, or zero) on each of these five pieces of the domain.

---

4. A function's derivative $f'(x)$ is negative for $x < 2$, zero at $x = 2$, and positive for $x > 2$. Describe in words what the original graph of f is doing at and around $x = 2$.

PART C · DIFFERENTIABILITY AND CONTINUITY

5. Let $g(x) = |x - 3|$. Determine whether g is differentiable at $x = 3$. Show the one-sided difference quotients and state your conclusion, including whether g is continuous at $x = 3$.

PART D · EXPLAIN YOUR THINKING

6. Explain in words why every differentiable function must be continuous, but not every continuous function is differentiable. Support your explanation with one specific example from this problem set or from Lesson 4.04.